
Homework # 1 Solution

1. A certain fabrication process produces 75% chips that are good (i.e., actually defect free). The testing mechanism for the finished ICs has an accuracy of 96% (i.e., probability that a good chip passes the test). Find the probability that a chip passes the test, and the defect-level of this test process for producing ICs.

The following convention can be used:

PQ: chip is good

FQ: chip is bad

P: chip passes the test

F: chip fails the test

A 75% good chip production means, $\text{Prob}(PQ) = 0.75$ and $\text{Prob}(FQ) = 0.25$. Similarly, from the given data, we can see that $\text{Prob}(P/PQ) = 0.96$ and $\text{Prob}(P/FQ) = 0.04$.

We have to calculate the probability of “passing”, $\text{Prob}(P)$.

$$\begin{aligned}\text{Prob}(P) &= \text{Prob}(P/PQ) \times \text{Prob}(PQ) + \text{Prob}(P/FQ) \times \text{Prob}(FQ) \\ &= 0.96 \times 0.75 + 0.04 \times 0.25 = 0.73\end{aligned}$$

$$\begin{aligned}\text{Defect level} &= \frac{\text{Bad chips that pass tests}}{\text{All chips that pass tests}} = \text{Prob}(FQ|P) = \frac{\text{Prob}(P/FQ)\text{Prob}(FQ)}{\text{Prob}(P)} \\ &= \frac{0.04 \times 0.25}{0.73} = 0.013699\end{aligned}$$

The defect level is 13, 699 ppm (parts per million) . The yield is estimated to be 0.73 by estimating the probability of chips that pass the test.

2. A mixed-signal VLSI chip has 1024 digital pins. To reduce the testing cost, it is decided to design the digital part with *built-in self-test* (BIST) circuitry. Thus, most of the digital signals can be generated or analyzed on the chip and do not have to be supplied from/to the ATE. It can now be tested on a 256-pin ATE. Calculate the per second cost of testing.

Note: Use the following assumptions:

ATE cost: \$1.2M + \$3000/pin,

ATE depreciation rate: 20% per year,

ATE maintenance cost: 2% of purchase price per year,

Operating cost (of building, facilities, auxiliary equipment (wafer and chip handler, fixtures, etc.) and personnel): \$0.5M per year.

Following Example 1.2 of the BA book (pp. 10-11), we obtain

$$\text{ATE purchase price} = \$1.2\text{M} + 256 \times \$3,000 = \$1.968\text{M}$$

Assuming a 20% per year linear rate of depreciation, a maintenance cost of 2% of the

price, and an annual operating cost of \$0.5M,

$$\text{Running cost} = \$1.968\text{M} \times 0.2 + \$1.968\text{M} \times 0.02 + \$0.5\text{M} = \$932,960/\text{year}$$

$$\text{Testing cost} = \frac{\$932,960}{365 \times 24 \times 3600} = 2.96 \text{ cents/second}$$

Testing cost is same as running cost calculated per second.

Testing cost of the self-test design is 2.96 cents per second, down from 4.50 cents per second calculated in Example 1.2 in the BA book for 1024-pin alternative chip without BIST.

3. A digital system board uses 64 ICs manufactured in two different independent processes. 48 of the ICs are manufactured in process A with 86% yield and with 96% fault coverage. The remaining ICs (16) are manufactured in process B with 72% yield and with 98% fault coverage. Assuming that the production process and the test program which is used to identify faulty boards are perfect, what is the quality level of the manufactured board? Assume quality level QL is the likelihood that all ICs are fault free.

Note: You will have to choose appropriate yield models based on the provided information to compute the defect levels and specify the yield models that are used.

Based on the provided information, the only yield model which can be used is the Brown & Williams: $DL = 1 - Y^{1-FC}$.

For ICs with Process A:

Fault coverage = 96%

Yield = 86%

$$DL1 = (\text{DefectLevel1}) = 1 - Y^{1-FC} = 0.00601475$$

For ICs with Process B:

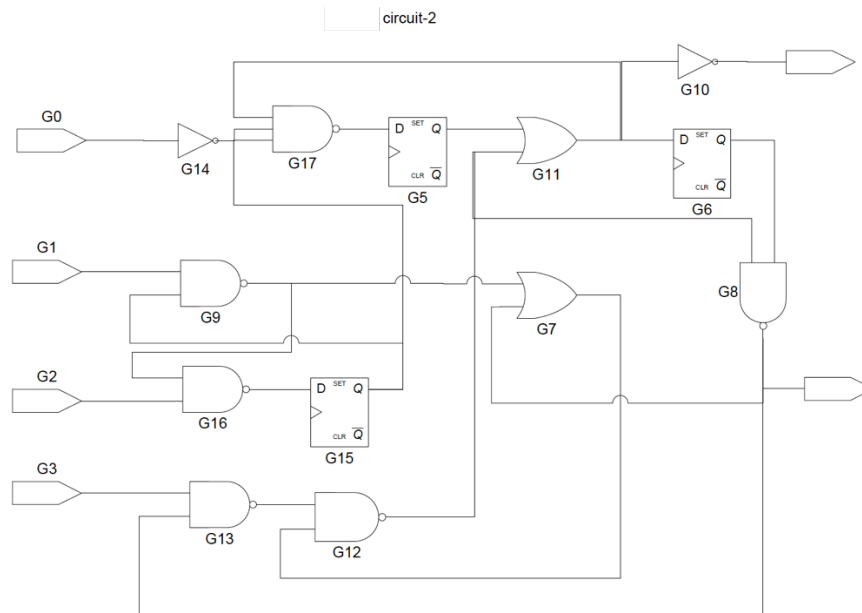
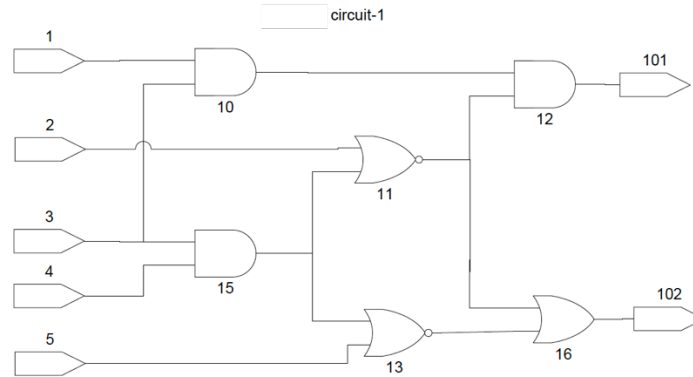
Fault coverage = 98%

Yield = 72%

$$DL2 = (\text{DefectLevel2}) = 1 - Y^{1-FC} = 0.006548545$$

$$\text{Board Quality Level} = \text{Prob}(\text{all IC's are fault free}) = (1 - DL1)^{48}(1 - DL2)^{16} = 0.6738825$$

4. Draw the two circuits corresponding to the files circuit-1 and circuit-2. These two files are linked from the Canvas main page under the material for HW1.



5. Using your circuit drawn for *circuit-1* in problem 4, simulate the following four vectors and specify the output of each vector.

						PO	PO
PI	1	2	3	4	5	101	102
V1	0	0	1	0	0	0	1
V2	U	0	0	0	0	0	1
V3	0	1	0	U	U	0	U
V4	1	U	0	0	1	0	U